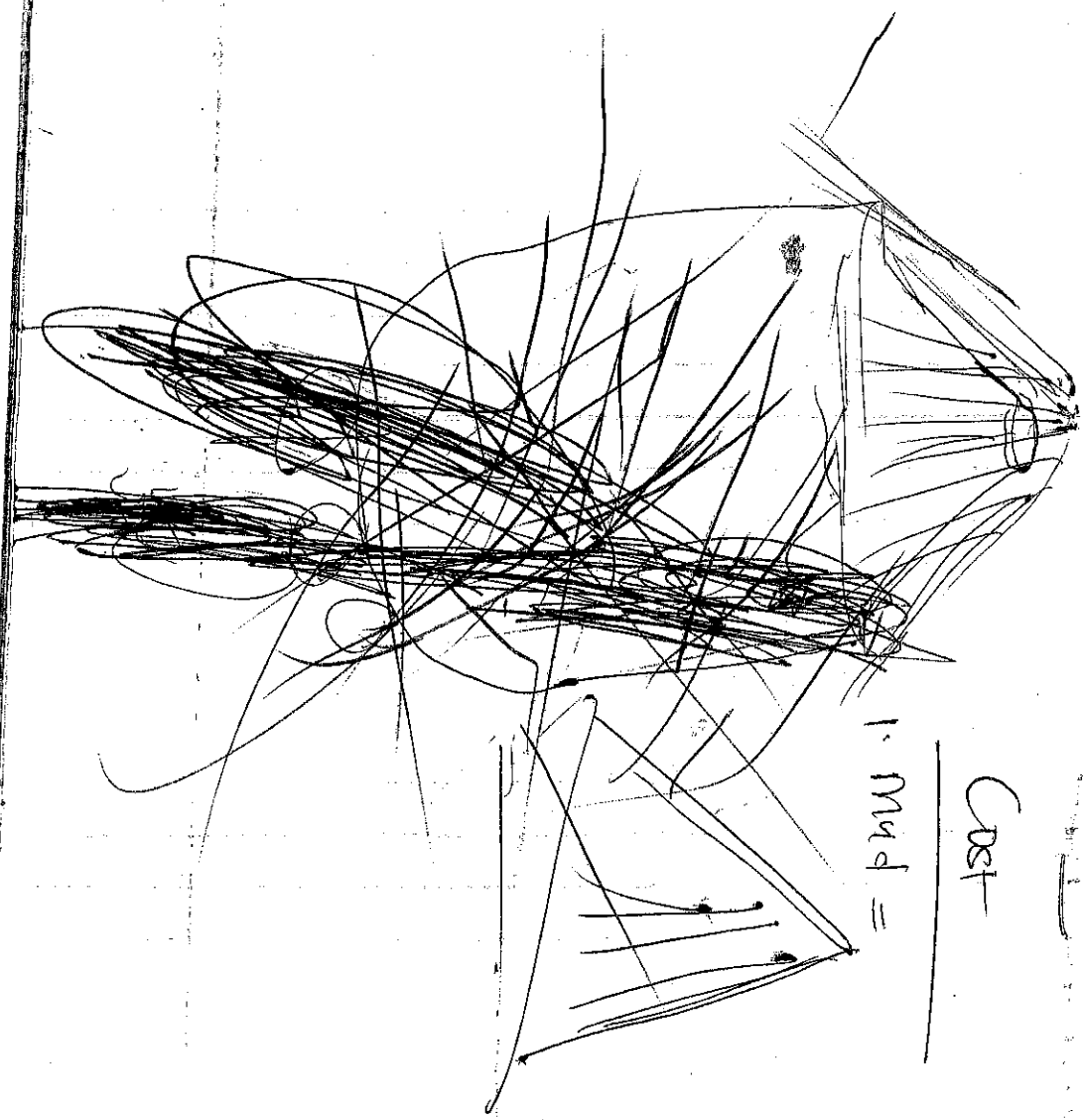




Cost = 1. Mnd =

* Native and Types of damage to Agric. produce

It is estimated that 5-10% of the world food production is damaged by insects during storage. Insects feed on the germ and endosperm causing losses in weight as well as nutrient. Beside the cause contamination with their excreta and death bodies.

Damages can be grouped into:

- (i) Direct damages.
- (ii) Indirect damages.

Direct Damages

The direct damages to agricultural produce in stores are as follows:-

1. Some insect consume germ, some endosperm and others eat away both. The result is losses of weight, losses or conversion of nutrients, loss of germination power, loss in gradation and consequently fall in market value.
2. Contamination may be with the debris

3. Bodies, cast skin, excreta, Obnoxious odour and or webbing, Structures and Containers may also be damaged by causing tunnelling in wooden parts resulting in the weakening of the structure or Container which also affects the produce stored in it.

Indirect damages

The indirect agricultural damages in stores are as follows:-

1. It may create nesting and migration of moisture
2. It may create distribution of parasite to man, certain type worms use stored grains insect as intermediate host
3. It causes consumer resistance / rejection which may lower the prestige or value of the product

Q.0

Causes of damages in stores

Insect-pest form one of the most important factor responsible for damages in agricultural production at various stages.

1. Living Organisms - living organisms and environment interact to bring about spoilage of stored products. Living organisms may be plants, insect, pest, man, animal, bacteria, fungi etc.
2. Environmental conditions - The environmental conditions are such as moisture, temperature and sanitation level etc can cause damage in stores.

Sources of Infestation of stored Produce

There are mainly five sources of infestation/infestation of stored produce namely -

1. field itself.
2. ^{infested} gunnies (bag or package)
3. ~~infested~~ transport
4. " godown (store or shade)
5. ~~infested~~ ^{infested} stocks

(1) field
Insect may attack the crop in the field itself. Grains effected by this insect when brought to the storage centres will be subjected to their continuous attack. The infestation may be visible or invisible. The stock should, therefore be fumigated immediately on receipt.

2. Infested gunnies
Even if sound and new gunnies are used for packing the newly harvested grain, the insect-hiding in the seams of the gunnies will attack the freshly harvested stocks. The gunnies should be fumigated before packing the freshly ~~that~~ harvested grains.

3. Infested transport
The transport used for carrying the new harvested stock should be invariably sprayed DDP or other malathion. If the transport like bullock cart or tractor ^{trailing} has been used for carrying infested stock on the previous occasions, the left over insect may attack the stocks now loaded into them.

4. Infested godown

Even if the stocks are moved to a freshly tracing over godown, it is

crevices
like small
holes.

not free from troubles. The insect
present in the ~~test~~^{cracks} and may
emerge out and attack the grains.
Hence the cracks and the crevices
should ^{be} plugged and empty
godowns should be ^{thoroughly} ~~thoroughly~~ ^{clean}
and fumigated.

5. Infested Stocks

In case stock ~~stocks~~ are brought
to a godown where infested stocks
are in storage cross infestation
takes place.

Types of Insects

Insect can be grouped into the
following groups.

1. Major Insects
2. Minor Insects
3. Incidental Insects
4. Predators and Parasites

①

Major Insects

Are those which attack the ^{item} food grains
directly and destroyed completely.
Examples are Khapra beetle, Ragee
Rhizopertha sp and Setopholius sp.
etc.

②

Minor Insect

Are those which follow the destruction
already started by the major insect
but cannot directly attack the grain
and cause the destruction independ-
ently. Examples are Laemophloeus
minutus and Oryzaephilus surin-
amensis. These insect are associated
with particular environmental conditions
such as moisture, temperature, Sani-
tation level etc.

③

Incidental Insects, predicted

There are certain un-determined species
of insect which may or may not
harm stored food grains, these are called
incidental insects which contaminate the

grains with the presence.

4. Predators or Parasites

The insect which depend on prey on the above type of insect are called parasite and predators respectively.

BIOLOGY OF PEST & PATHOGEN IN STORES -

Common insect of stored grains. Some of the insect general found in storage of food grain, and other agricultural commodity are described in brief below.

- ✓ 1. Sitophilus Oryzae and S. granarius
They are called as rice weevil and grainy weevil. The adult can easily be identify by the snout-like projection or proboscis from the head. The larvae are white tiny regular masses no definite shape. The two spp can be distinguished by the dense round punctures in the

thorax in rice weevil and less dense elongated pits in grainy weevil both adult and larval stage are harmful.

2. Rhizopertha dominica - This is called lesser grain borer and can be identify by the cylindrical body of the insect with the head bend down ward and its shows movement. The larvae are uniform in shape both adult and larvae are harmful.

- ✓ 3. Proga domus granaria - The general name is Khapra. The adult are brownish oval body. flies having a very short span of life. The larvae of this insect is harmful. The larvae have many segmented and yellowish brown in colour with movable erect hair with the posterior segment forming a tail or a tuft (bunch of hair).

✓ 4. Lasioderma Semicorne - These are commonly known as cigar beetles or tobacco box beetles affecting tobacco (They bore the stem) chilly pepper, ginger etc both leaves and adults are harmful. The adult can be identified by their yellowish brown colour and the hemispherical shape quite different from Knapers with head hidden under the domed-neck shield. The freshly emerged larvae are very mobile.

5. Thiodium castaneum and Thiodium Confusum. The common name are red flower beetles and confused flour beetle. They can be noticed everywhere in association with other insect. The adults are red in colour with black (active) movement. The confused beetles can be distinguished from the other spp by examining their antennae.

The red flower beetles the segment of the antennae are smaller at the base and abruptly larger at the apex where as in the confused beetles, the size of the segment gradually increases from the base to the apex.

6. Leptothorax Orizae. They are commonly known as long head flour beetle. The adults are cylinder flattened insect, pale yellow in colour, the antennae seven segmented club shape and small.

7. Laemophloeus minutus - These are called flat small size brick movement bright brown colour and the longer antennae. They can also be seen in association with Orizaphilus and Thiodium.

✓ 8. Bruchus Spp. they are commonly known as pulse beetle. The adult always have a peritreme and almost

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triangular shape brownish grey in colour covered with bristles.

(7) Stegoderm Paniceum : These are commonly known as spice beetle, the adult have cylindrical shape brown in colour. They can be distinguished from Lessiodermis serricornis by the antennae which are longer and the last 3 segments which clubbed.

(10) Corcyra Cephalonica : This is a moth, Pearl green in colour with a wing spread 12-14mm. They can be distinguished by a closed straight peak formed by the palpi pointing forward and antennae not crossed over wings while the insect is on rest. Larva are white in colour with light brownish yellow broad head. It is commonly known as rice moth.

✓
(11) Ephestia cautella : It is also commonly known as fig moth or almond moth, greyish in colour with a wing spread of about 20mm. They can be distinguished from the transverse stripes on the outer wings, the larva are pinkish in colour.

(12) Plodia interpunctella : The common name is Indian meal moth. The adults have wings spread of 14-20mm. When at rest with close it is 8-10mm long and greyish yellow in colour. Larva are yellowish white, some are reddish or even greenish with a brown head.

RATS

Rats have to be found to be taking ~~to~~ be taking a lion share in the devastation of food materials. It has been estimated that one rat consumes 10kg of grain in a year and contaminate more than 10 times what it actually eats, with its urine and excreta. One rat can eat 25-150

Pellets and 10-20 millilitre urine a day, besides shedding its plenty of hair. Their multiplication is also ~~also~~ so tremendous that the rat population outpace the human population by 6 times. Rats also spread fatal diseases like Plague, jaundice, rat bite fever, typhus etc. There are 118 species of rats, house mouse, brown rat, Indian mole rat, large bandicoot rat, short-tailed rat, Indian field mouse, Indian gerbil or sunset rat etc.

CONTROL METHODS.

1. Rat control measure: Sanitation in houses and godowns is significantly helpful in reducing the rat menace. Rats multiply rapidly if they get food and shelter. Waste grains should be removed, grain sacks should be removed, grain sacks should be stored about 50 cm above the floor. And the grain should be protected with concrete, wire,

or sheet metal. In order to avoid the interest of rat in houses or godowns the plate should be fitted to the doors and windows at the bottom in such a way that once they are closed, no space is left for making entry by the rat. In construction of rat proof godowns, the following things are to be given due consideration:-

1. The foundation is kept deep and the plinth is constructed about 1m above the ground level in L shape having collar projected from all the side.
2. A godown should not have provision of steps.
3. fitting of wire ~~netting~~ to the drain pipes.
4. If corrugated sheet or ^{asbestos} sheet is used for ceiling, the space b/w the wall and the ceiling should also be covered with wire ~~netting~~.

The chemical used to kill rat and mice are called rodenticides. Rodenticides are usually mixed with baits such as

Stop

fresh, meat, rice, pepper, butter etc. The baits must be mixed so directly & as to little may not poison. Apart from the above method, rat traps are also used to control rat menace.

45 Rat Control Measures:

- 1) Stores should be always kept clean and be fumigated by packing with any products.
- 2) Waste & broken kernels should always be removed.
- 3) Grain stores should be kept 20cm above the floor level.
- 4) chemical like rodenticides should be used in controlling rats, but should be mixed very well according to the recommendations, & as to little may not poison them.
- 5) Rat traps are also used to control rat menace.

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When the produce in stores has been attacked by any insect-pest, you notice a lot of physical changes to the produce stored. Grains will undergo color formation, unpleasant smell and odour, broken kernels, their fleeces & husk, or the cast-bodies etc.

As the result of action of their damage - on experience loss in weight, germ %, fall in market value.

S9 Outline the various insects control methods you know.

1. Hygienic method of insect & pest control
2. Hermetic technology of insect & pest control
3. Aeration or degeration technology
4. Heat-treatment-methods
5. Use of chemicals

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Define aeration and enumerate its benefit in the storage of grains.

Aeration can be defined as process of moving air through stored grain at low flow rate to maintain or improve its quality.

The benefits are

1. It coats the grains & slows down insect-activities by coating the grains, & also prolongs the effectiveness of pesticides or insecticides.

②. It can provide an appreciable drying function.